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Live demo: archlens-xhxr.onrender.com · Code: github.com/vishnuatgit/ArchLens

ArchLens

Scores any public GitHub repository from 0 to 100 across seven engineering dimensions, as a web app, a REST API and a CLI.

YEAR

2026

BUILT WITH

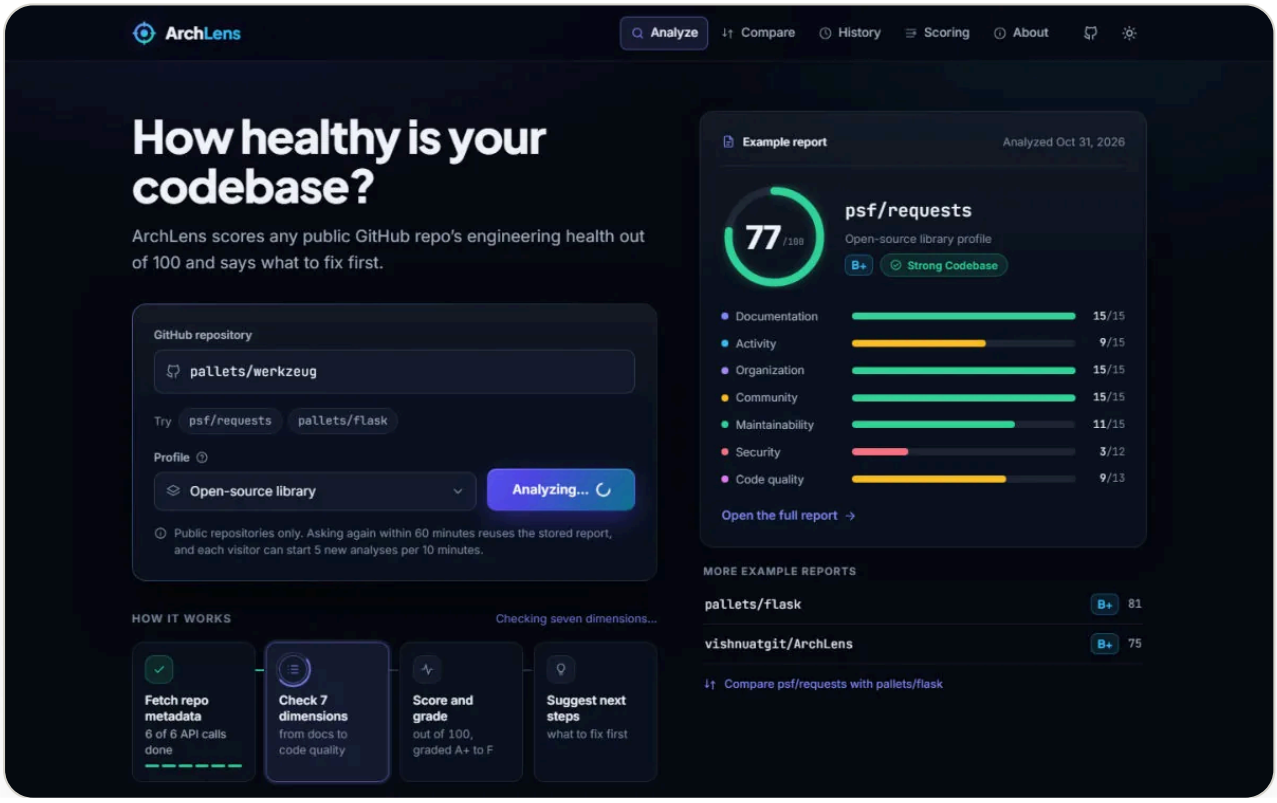
FastAPI · SQLAlchemy · Docker

PROOF

- 266 tests · CI passing

SOURCE

github.com/vishnuatgit/ArchLens



(01) The problem

WHY IT EXISTS

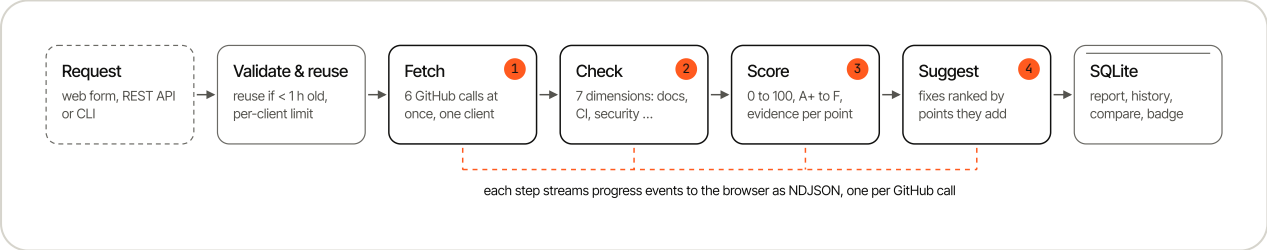
ArchLens checks a repository's *engineering health* in one request.

By hand, you would open the README, look for a license, tests and CI, and scan recent commits and issues. ArchLens does those checks at once, shows what it found for each, and lists what is missing as fixes, most valuable first.

It reads repository metadata through the GitHub REST API and never clones the code, so an analysis takes six read-only API calls. Every new analysis is stored in SQLite, so reports can be revisited, compared and shared.

(02) How it works

FROM THE README



1 **Fetch repo metadata.** Repository stats, the root file listing, languages, contributors, commits from the last 30 days and the workflow folder, read with one shared `httpx.AsyncClient` and `asyncio.gather`. If one call fails, the others are cancelled.

2 **Check 7 dimensions.** Documentation, activity, organization, community, maintainability, security and code quality: a README, license, test folder, CI workflows, lockfile, `SECURITY.md`, linter and type-checker config, recent commits and open issues.

3 **Score and grade.** Points add up to 100 and map to a grade from A+ (95 and above) to F (below 35) and a health tier, with the evidence behind every point.

4 **Suggest next steps.** Each failed check becomes a fix with the points it would add, ranked, and exportable as a Markdown checklist.

(03) Key decisions

5 CHOICES

01 One shared async client, metadata only

The six GitHub calls run concurrently through one `httpx.AsyncClient` shared for the app's lifetime. If one fails (a 404 on the metadata call, say), the calls still running are cancelled instead of spending more GitHub quota.

02 Upstream errors become clear responses

One error shape across the web app, the API and the CLI: a missing repository is a `404 REPO_NOT_FOUND`, a used-up GitHub quota a `503 RATE_LIMIT_EXCEEDED`, an unreachable GitHub a `502 GITHUB_UNAVAILABLE`, and too many analyses a `429` with `Retry-After`.

03 Cache and rate-limit for a free host

A report of the same repository and profile from the last hour is reused, and each client can start 5 new analyses per 10 minutes. An analysis that fails because GitHub is down or out of quota is given back.

04 Progress from real events

`POST /api/v1/analyze/stream` streams the pipeline's progress as newline-delimited JSON, one event per GitHub call, so the steps on the home page light up from what the server is actually doing, and a failed step says why.

05 Scoring you can check

The `/scoring` page is generated from the scorer's own constants, and tests run the real scorer against it, so the explanation cannot drift from the code.

(04) Proof

TESTS · CI · STACK

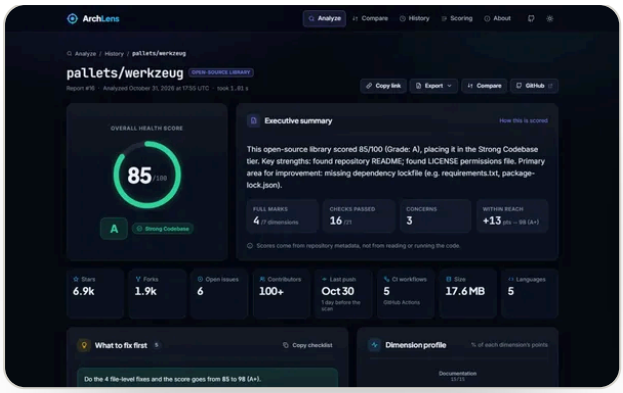
266	Passing	6
PYTEST TESTS, GITHUB MOCKED, NO NETWORK	CI ON GITHUB ACTIONS: PYTEST, RUFF, BLACK	GITHUB API CALLS PER ANALYSIS, RUN CONCURRENTLY

Unit tests cover the scorer, the scoring page, the evidence (the real scorer over about 900 generated repositories), the fix plan, the progress events, the GitHub client and the rate limiter. Integration tests drive every page and API endpoint, including the progress stream, through FastAPI's `TestClient`. CI runs ↗

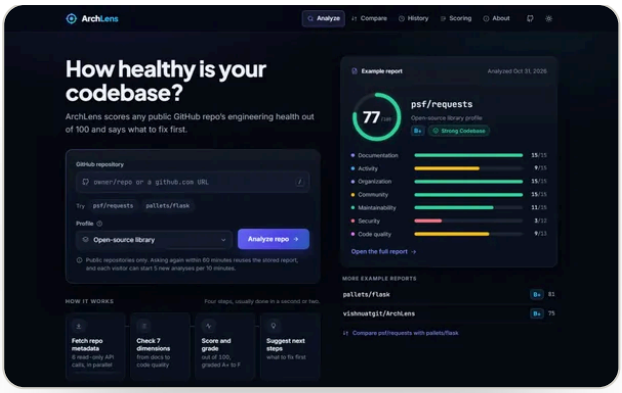
`Python 3.11+` `FastAPI` `httpx` `SQLAlchemy 2` `Alembic` `SQLite` `Jinja2` `Typer + Rich` `pytest` `Ruff` `Black` `Docker` `GitHub Actions`

(05) Screenshots

FROM THE REPOSITORY



A report: the score and grade, the summary, and the facts behind them.



The home page: the steps light up as the server streams its progress.

(06) Limitations

HONESTLY

— **Metadata only.** It reads repository statistics and the root file listing. It does not read source files, run tests or measure coverage.

— **Simple rules.** A file earns points just by existing, whatever its quality, and stars say little about code health. Treat a low score as a list of things to check.

— **Capped counts.** Contributors and recent commits come from one page of GitHub results, so they stop at 100.

— **Single instance.** The rate limiter lives in memory and resets on restart.

— **Ephemeral demo storage.** On the free host, reports other than the bundled examples disappear when the service restarts.

NEXT CASE STUDY

DeepBell ↗

A mini SRE lab in FastAPI: it detects anomalies in its own CPU and memory, writes incident reports and restores from snapshots.